C:\Users\minhq\AppData\Local\Temp\wireshark_Adapter for loopback traffic captureI69SU3.pcapng 22 total packets, 18 shown

```
No.     Time           Source                Destination           Protocol Length Info
      1 0.000000000    127.0.0.1             127.0.0.1             TCP      56     55454 → 5001 [SYN] Seq=0 Win=3 Len=0 MSS=65495
WS=1 SACK_PERM
Frame 1: Packet, 56 bytes on wire (448 bits), 56 bytes captured (448 bits) on interface \Device\NPF_Loopback, id 0
Null/Loopback
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 55454, Dst Port: 5001, Seq: 0, Len: 0
    Source Port: 55454
    Destination Port: 5001
    [Stream index: 0]
    [Stream Packet Number: 1]
    [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 0]
    Sequence Number: 0    (relative sequence number)
    Sequence Number (raw): 992947609
    [Next Sequence Number: 1    (relative sequence number)]
    Acknowledgment Number: 0
    Acknowledgment number (raw): 0
    1000 .... = Header Length: 32 bytes (8)
    Flags: 0x002 (SYN)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Accurate ECN: Not set
        .... 0... .... = Congestion Window Reduced: Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...0 .... = Acknowledgment: Not set
        .... .... 0... = Push: Not set
        .... .... .0.. = Reset: Not set
        .... .... ..1. = Syn: Set
            [Expert Info (Chat/Sequence): Connection establish request (SYN): server port 5001]
                [Connection establish request (SYN): server port 5001]
                [Severity level: Chat]
                [Group: Sequence]
        .... .... ...0 = Fin: Not set
        [TCP Flags: ··········S·]
    Window: 3
    [Calculated window size: 3]
    Checksum: 0x21ff [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    Options: (12 bytes), Maximum segment size, No-Operation (NOP), Window scale, No-Operation (NOP), No-Operation (NOP), SACK
permitted
        TCP Option - Maximum segment size: 65495 bytes
        TCP Option - No-Operation (NOP)
        TCP Option - Window scale: 0 (multiply by 1)
        TCP Option - No-Operation (NOP)
        TCP Option - No-Operation (NOP)
        TCP Option - SACK permitted
    [Timestamps]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
No.     Time           Source                Destination           Protocol Length Info
      2 0.000068900    127.0.0.1             127.0.0.1             TCP      56     5001 → 55454 [SYN, ACK] Seq=0 Ack=1 Win=4 Len=0
MSS=65495 WS=1 SACK_PERM
Frame 2: Packet, 56 bytes on wire (448 bits), 56 bytes captured (448 bits) on interface \Device\NPF_Loopback, id 0
Null/Loopback
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 5001, Dst Port: 55454, Seq: 0, Ack: 1, Len: 0
    Source Port: 5001
    Destination Port: 55454
    [Stream index: 0]
    [Stream Packet Number: 2]
    [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 0]
    Sequence Number: 0    (relative sequence number)
    Sequence Number (raw): 2008480766
    [Next Sequence Number: 1    (relative sequence number)]
    Acknowledgment Number: 1    (relative ack number)
    Acknowledgment number (raw): 992947610
    1000 .... = Header Length: 32 bytes (8)
    Flags: 0x012 (SYN, ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Accurate ECN: Not set
        .... 0... .... = Congestion Window Reduced: Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...1 .... = Acknowledgment: Set
```

```
        .... .... 0... = Push: Not set
        .... .... .0.. = Reset: Not set
        .... .... ..1. = Syn: Set
            [Expert Info (Chat/Sequence): Connection establish acknowledge (SYN+ACK): server port 5001]
                [Connection establish acknowledge (SYN+ACK): server port 5001]
                [Severity level: Chat]
                [Group: Sequence]
        .... .... ...0 = Fin: Not set
        [TCP Flags: ·····A··S·]
    Window: 4
    [Calculated window size: 4]
    Checksum: 0xae37 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    Options: (12 bytes), Maximum segment size, No-Operation (NOP), Window scale, No-Operation (NOP), No-Operation (NOP), SACK
permitted
        TCP Option - Maximum segment size: 65495 bytes
        TCP Option - No-Operation (NOP)
        TCP Option - Window scale: 0 (multiply by 1)
        TCP Option - No-Operation (NOP)
        TCP Option - No-Operation (NOP)
        TCP Option - SACK permitted
    [Timestamps]
    [SEQ/ACK analysis]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
No.     Time           Source                Destination           Protocol Length Info
      3 0.000103100    127.0.0.1             127.0.0.1             TCP      44     55454 → 5001 [ACK] Seq=1 Ack=1 Win=3 Len=0
Frame 3: Packet, 44 bytes on wire (352 bits), 44 bytes captured (352 bits) on interface \Device\NPF_Loopback, id 0
Null/Loopback
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 55454, Dst Port: 5001, Seq: 1, Ack: 1, Len: 0
    Source Port: 55454
    Destination Port: 5001
    [Stream index: 0]
    [Stream Packet Number: 3]
    [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 0]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 992947610
    [Next Sequence Number: 1    (relative sequence number)]
    Acknowledgment Number: 1    (relative ack number)
    Acknowledgment number (raw): 2008480767
    0101 .... = Header Length: 20 bytes (5)
    Flags: 0x010 (ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Accurate ECN: Not set
        .... 0... .... = Congestion Window Reduced: Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...1 .... = Acknowledgment: Set
        .... .... 0... = Push: Not set
        .... .... .0.. = Reset: Not set
        .... .... ..0. = Syn: Not set
        .... .... ...0 = Fin: Not set
        [TCP Flags: ·····A····]
    Window: 3
    [Calculated window size: 3]
    [Window size scaling factor: 1]
    Checksum: 0xe927 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    [Timestamps]
    [SEQ/ACK analysis]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
No.     Time           Source                Destination           Protocol Length Info
      4 0.000173500    127.0.0.1             127.0.0.1             TCP      48     [TCP Window Full] 55454 → 5001 [ACK] Seq=1 Ack=1
Win=3 Len=4
Frame 4: Packet, 48 bytes on wire (384 bits), 48 bytes captured (384 bits) on interface \Device\NPF_Loopback, id 0
Null/Loopback
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 55454, Dst Port: 5001, Seq: 1, Ack: 1, Len: 4
    Source Port: 55454
    Destination Port: 5001
    [Stream index: 0]
    [Stream Packet Number: 4]
```

```
    [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 4]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 992947610
    [Next Sequence Number: 5    (relative sequence number)]
    Acknowledgment Number: 1    (relative ack number)
    Acknowledgment number (raw): 2008480767
    0101 .... = Header Length: 20 bytes (5)
    Flags: 0x010 (ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Accurate ECN: Not set
        .... 0... .... = Congestion Window Reduced: Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...1 .... = Acknowledgment: Set
        .... .... 0... = Push: Not set
        .... .... .0.. = Reset: Not set
        .... .... ..0. = Syn: Not set
        .... .... ...0 = Fin: Not set
        [TCP Flags: ·····A····]
    Window: 3
    [Calculated window size: 3]
    [Window size scaling factor: 1]
    Checksum: 0x3250 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    [Timestamps]
    [SEQ/ACK analysis]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
    TCP payload (4 bytes)
Data (4 bytes)
0000  47 6f 6f 64                                        Good
No.     Time           Source                Destination           Protocol Length Info
      5 0.000187200    127.0.0.1             127.0.0.1             TCP      44     5001 → 55454 [ACK] Seq=1 Ack=5 Win=4 Len=0
Frame 5: Packet, 44 bytes on wire (352 bits), 44 bytes captured (352 bits) on interface \Device\NPF_Loopback, id 0
Null/Loopback
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 5001, Dst Port: 55454, Seq: 1, Ack: 5, Len: 0
    Source Port: 5001
    Destination Port: 55454
    [Stream index: 0]
    [Stream Packet Number: 5]
    [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 0]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 2008480767
    [Next Sequence Number: 1    (relative sequence number)]
    Acknowledgment Number: 5    (relative ack number)
    Acknowledgment number (raw): 992947614
    0101 .... = Header Length: 20 bytes (5)
    Flags: 0x010 (ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Accurate ECN: Not set
        .... 0... .... = Congestion Window Reduced: Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...1 .... = Acknowledgment: Set
        .... .... 0... = Push: Not set
        .... .... .0.. = Reset: Not set
        .... .... ..0. = Syn: Not set
        .... .... ...0 = Fin: Not set
        [TCP Flags: ·····A····]
    Window: 4
    [Calculated window size: 4]
    [Window size scaling factor: 1]
    Checksum: 0xe922 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    [Timestamps]
    [SEQ/ACK analysis]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
No.     Time           Source                Destination           Protocol Length Info
      6 0.000192900    127.0.0.1             127.0.0.1             TCP      48     [TCP Window Full] 55454 → 5001 [ACK] Seq=5 Ack=1
Win=3 Len=4
Frame 6: Packet, 48 bytes on wire (384 bits), 48 bytes captured (384 bits) on interface \Device\NPF_Loopback, id 0
```

```
  Null/Loopback
  Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
  Transmission Control Protocol, Src Port: 55454, Dst Port: 5001, Seq: 5, Ack: 1, Len: 4
      Source Port: 55454
      Destination Port: 5001
      [Stream index: 0]
      [Stream Packet Number: 6]
      [Conversation completeness: Complete, WITH_DATA (31)]
      [TCP Segment Len: 4]
      Sequence Number: 5    (relative sequence number)
      Sequence Number (raw): 992947614
      [Next Sequence Number: 9    (relative sequence number)]
      Acknowledgment Number: 1    (relative ack number)
      Acknowledgment number (raw): 2008480767
      0101 .... = Header Length: 20 bytes (5)
      Flags: 0x010 (ACK)
          000. .... .... = Reserved: Not set
          ...0 .... .... = Accurate ECN: Not set
          .... 0... .... = Congestion Window Reduced: Not set
          .... .0.. .... = ECN-Echo: Not set
          .... ..0. .... = Urgent: Not set
          .... ...1 .... = Acknowledgment: Set
          .... .... 0... = Push: Not set
          .... .... .0.. = Reset: Not set
          .... .... ..0. = Syn: Not set
          .... .... ...0 = Fin: Not set
          [TCP Flags: ·····A····]
      Window: 3
      [Calculated window size: 3]
      [Window size scaling factor: 1]
      Checksum: 0x5940 [unverified]
      [Checksum Status: Unverified]
      Urgent Pointer: 0
      [Timestamps]
      [SEQ/ACK analysis]
      [Client Contiguous Streams: 1]
      [Server Contiguous Streams: 1]
      TCP payload (4 bytes)
  Data (4 bytes)
  0000  20 6d 6f 72                                        mor
  No.     Time           Source                Destination           Protocol Length Info
        7 0.000199500    127.0.0.1             127.0.0.1             TCP      44     5001 → 55454 [ACK] Seq=1 Ack=9 Win=4 Len=0
  Frame 7: Packet, 44 bytes on wire (352 bits), 44 bytes captured (352 bits) on interface \Device\NPF_Loopback, id 0
  Null/Loopback
  Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
  Transmission Control Protocol, Src Port: 5001, Dst Port: 55454, Seq: 1, Ack: 9, Len: 0
      Source Port: 5001
      Destination Port: 55454
      [Stream index: 0]
      [Stream Packet Number: 7]
      [Conversation completeness: Complete, WITH_DATA (31)]
      [TCP Segment Len: 0]
      Sequence Number: 1    (relative sequence number)
      Sequence Number (raw): 2008480767
      [Next Sequence Number: 1    (relative sequence number)]
      Acknowledgment Number: 9    (relative ack number)
      Acknowledgment number (raw): 992947618
      0101 .... = Header Length: 20 bytes (5)
      Flags: 0x010 (ACK)
          000. .... .... = Reserved: Not set
          ...0 .... .... = Accurate ECN: Not set
          .... 0... .... = Congestion Window Reduced: Not set
          .... .0.. .... = ECN-Echo: Not set
          .... ..0. .... = Urgent: Not set
          .... ...1 .... = Acknowledgment: Set
          .... .... 0... = Push: Not set
          .... .... .0.. = Reset: Not set
          .... .... ..0. = Syn: Not set
          .... .... ...0 = Fin: Not set
          [TCP Flags: ·····A····]
      Window: 4
      [Calculated window size: 4]
      [Window size scaling factor: 1]
      Checksum: 0xe91e [unverified]
      [Checksum Status: Unverified]
      Urgent Pointer: 0
      [Timestamps]
```

```
    [SEQ/ACK analysis]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
No.     Time           Source                Destination           Protocol Length Info
      8 0.000204000    127.0.0.1             127.0.0.1             TCP      48     [TCP Window Full] 55454 → 5001 [PSH, ACK] Seq=9
Ack=1 Win=3 Len=4
Frame 8: Packet, 48 bytes on wire (384 bits), 48 bytes captured (384 bits) on interface \Device\NPF_Loopback, id 0
Null/Loopback
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 55454, Dst Port: 5001, Seq: 9, Ack: 1, Len: 4
    Source Port: 55454
    Destination Port: 5001
    [Stream index: 0]
    [Stream Packet Number: 8]
    [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 4]
    Sequence Number: 9    (relative sequence number)
    Sequence Number (raw): 992947618
    [Next Sequence Number: 13    (relative sequence number)]
    Acknowledgment Number: 1    (relative ack number)
    Acknowledgment number (raw): 2008480767
    0101 .... = Header Length: 20 bytes (5)
    Flags: 0x018 (PSH, ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Accurate ECN: Not set
        .... 0... .... = Congestion Window Reduced: Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...1 .... = Acknowledgment: Set
        .... .... 1... = Push: Set
        .... .... .0.. = Reset: Not set
        .... .... ..0. = Syn: Not set
        .... .... ...0 = Fin: Not set
        [TCP Flags: ·····AP···]
    Window: 3
    [Calculated window size: 3]
    [Window size scaling factor: 1]
    Checksum: 0x0c43 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    [Timestamps]
    [SEQ/ACK analysis]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
    TCP payload (4 bytes)
Data (4 bytes)
0000  6e 69 6e 67                                        ning
No.     Time           Source                Destination           Protocol Length Info
      9 0.000210500    127.0.0.1             127.0.0.1             TCP      44     [TCP ZeroWindow] 5001 → 55454 [ACK] Seq=1 Ack=13
Win=0 Len=0
Frame 9: Packet, 44 bytes on wire (352 bits), 44 bytes captured (352 bits) on interface \Device\NPF_Loopback, id 0
Null/Loopback
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 5001, Dst Port: 55454, Seq: 1, Ack: 13, Len: 0
    Source Port: 5001
    Destination Port: 55454
    [Stream index: 0]
    [Stream Packet Number: 9]
    [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 0]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 2008480767
    [Next Sequence Number: 1    (relative sequence number)]
    Acknowledgment Number: 13    (relative ack number)
    Acknowledgment number (raw): 992947622
    0101 .... = Header Length: 20 bytes (5)
    Flags: 0x010 (ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Accurate ECN: Not set
        .... 0... .... = Congestion Window Reduced: Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...1 .... = Acknowledgment: Set
        .... .... 0... = Push: Not set
        .... .... .0.. = Reset: Not set
        .... .... ..0. = Syn: Not set
        .... .... ...0 = Fin: Not set
```

```
        [TCP Flags: ·····A····]
    Window: 0
    [Calculated window size: 0]
    [Window size scaling factor: 1]
    Checksum: 0xe91e [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    [Timestamps]
    [SEQ/ACK analysis]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
No.     Time           Source                Destination           Protocol Length Info
     10 0.000286100    127.0.0.1             127.0.0.1             TCP      47     [TCP ZeroWindow] [TCP Window Full] 5001 → 55454
[ACK] Seq=1 Ack=13 Win=0 Len=3
Frame 10: Packet, 47 bytes on wire (376 bits), 47 bytes captured (376 bits) on interface \Device\NPF_Loopback, id 0
Null/Loopback
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 5001, Dst Port: 55454, Seq: 1, Ack: 13, Len: 3
    Source Port: 5001
    Destination Port: 55454
    [Stream index: 0]
    [Stream Packet Number: 10]
    [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 3]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 2008480767
    [Next Sequence Number: 4    (relative sequence number)]
    Acknowledgment Number: 13    (relative ack number)
    Acknowledgment number (raw): 992947622
    0101 .... = Header Length: 20 bytes (5)
    Flags: 0x010 (ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Accurate ECN: Not set
        .... 0... .... = Congestion Window Reduced: Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...1 .... = Acknowledgment: Set
        .... .... 0... = Push: Not set
        .... .... .0.. = Reset: Not set
        .... .... ..0. = Syn: Not set
        .... .... ...0 = Fin: Not set
        [TCP Flags: ·····A····]
    Window: 0
    [Calculated window size: 0]
    [Window size scaling factor: 1]
    Checksum: 0x34b6 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    [Timestamps]
    [SEQ/ACK analysis]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
    TCP payload (3 bytes)
Data (3 bytes)
0000  48 65 6c                                          Hel
No.     Time           Source                Destination           Protocol Length Info
     11 0.000311600    127.0.0.1             127.0.0.1             TCP      44     55454 → 5001 [ACK] Seq=13 Ack=4 Win=3 Len=0
Frame 11: Packet, 44 bytes on wire (352 bits), 44 bytes captured (352 bits) on interface \Device\NPF_Loopback, id 0
Null/Loopback
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 55454, Dst Port: 5001, Seq: 13, Ack: 4, Len: 0
    Source Port: 55454
    Destination Port: 5001
    [Stream index: 0]
    [Stream Packet Number: 11]
    [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 0]
    Sequence Number: 13    (relative sequence number)
    Sequence Number (raw): 992947622
    [Next Sequence Number: 13    (relative sequence number)]
    Acknowledgment Number: 4    (relative ack number)
    Acknowledgment number (raw): 2008480770
    0101 .... = Header Length: 20 bytes (5)
    Flags: 0x010 (ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Accurate ECN: Not set
        .... 0... .... = Congestion Window Reduced: Not set
```

```
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...1 .... = Acknowledgment: Set
        .... .... 0... = Push: Not set
        .... .... .0.. = Reset: Not set
        .... .... ..0. = Syn: Not set
        .... .... ...0 = Fin: Not set
        [TCP Flags: ·····A····]
    Window: 3
    [Calculated window size: 3]
    [Window size scaling factor: 1]
    Checksum: 0xe918 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    [Timestamps]
    [SEQ/ACK analysis]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
No.     Time           Source                Destination           Protocol Length Info
     12 0.000319000    127.0.0.1             127.0.0.1             TCP      46     [TCP ZeroWindow] 5001 → 55454 [PSH, ACK] Seq=4
Ack=13 Win=0 Len=2
Frame 12: Packet, 46 bytes on wire (368 bits), 46 bytes captured (368 bits) on interface \Device\NPF_Loopback, id 0
Null/Loopback
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 5001, Dst Port: 55454, Seq: 4, Ack: 13, Len: 2
    Source Port: 5001
    Destination Port: 55454
    [Stream index: 0]
    [Stream Packet Number: 12]
    [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 2]
    Sequence Number: 4    (relative sequence number)
    Sequence Number (raw): 2008480770
    [Next Sequence Number: 6    (relative sequence number)]
    Acknowledgment Number: 13    (relative ack number)
    Acknowledgment number (raw): 992947622
    0101 .... = Header Length: 20 bytes (5)
    Flags: 0x018 (PSH, ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Accurate ECN: Not set
        .... 0... .... = Congestion Window Reduced: Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...1 .... = Acknowledgment: Set
        .... .... 1... = Push: Set
        .... .... .0.. = Reset: Not set
        .... .... ..0. = Syn: Not set
        .... .... ...0 = Fin: Not set
        [TCP Flags: ·····AP···]
    Window: 0
    [Calculated window size: 0]
    [Window size scaling factor: 1]
    Checksum: 0x7ca2 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    [Timestamps]
    [SEQ/ACK analysis]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
    TCP payload (2 bytes)
Data (2 bytes)
0000  6c 6f                                             lo
No.     Time           Source                Destination           Protocol Length Info
     13 0.000327000    127.0.0.1             127.0.0.1             TCP      44     55454 → 5001 [ACK] Seq=13 Ack=6 Win=3 Len=0
Frame 13: Packet, 44 bytes on wire (352 bits), 44 bytes captured (352 bits) on interface \Device\NPF_Loopback, id 0
Null/Loopback
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 55454, Dst Port: 5001, Seq: 13, Ack: 6, Len: 0
    Source Port: 55454
    Destination Port: 5001
    [Stream index: 0]
    [Stream Packet Number: 13]
    [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 0]
    Sequence Number: 13    (relative sequence number)
    Sequence Number (raw): 992947622
    [Next Sequence Number: 13    (relative sequence number)]
```

```
    Acknowledgment Number: 6    (relative ack number)
    Acknowledgment number (raw): 2008480772
    0101 .... = Header Length: 20 bytes (5)
    Flags: 0x010 (ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Accurate ECN: Not set
        .... 0... .... = Congestion Window Reduced: Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...1 .... = Acknowledgment: Set
        .... .... 0... = Push: Not set
        .... .... .0.. = Reset: Not set
        .... .... ..0. = Syn: Not set
        .... .... ...0 = Fin: Not set
        [TCP Flags: ·····A····]
    Window: 3
    [Calculated window size: 3]
    [Window size scaling factor: 1]
    Checksum: 0xe916 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    [Timestamps]
    [SEQ/ACK analysis]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
No.     Time           Source                Destination           Protocol Length Info
     14 0.000360600    127.0.0.1             127.0.0.1             TCP      44     [TCP Window Update] 5001 → 55454 [ACK] Seq=6
Ack=13 Win=4 Len=0
Frame 14: Packet, 44 bytes on wire (352 bits), 44 bytes captured (352 bits) on interface \Device\NPF_Loopback, id 0
Null/Loopback
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 5001, Dst Port: 55454, Seq: 6, Ack: 13, Len: 0
    Source Port: 5001
    Destination Port: 55454
    [Stream index: 0]
    [Stream Packet Number: 14]
    [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 0]
    Sequence Number: 6    (relative sequence number)
    Sequence Number (raw): 2008480772
    [Next Sequence Number: 6    (relative sequence number)]
    Acknowledgment Number: 13    (relative ack number)
    Acknowledgment number (raw): 992947622
    0101 .... = Header Length: 20 bytes (5)
    Flags: 0x010 (ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Accurate ECN: Not set
        .... 0... .... = Congestion Window Reduced: Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...1 .... = Acknowledgment: Set
        .... .... 0... = Push: Not set
        .... .... .0.. = Reset: Not set
        .... .... ..0. = Syn: Not set
        .... .... ...0 = Fin: Not set
        [TCP Flags: ·····A····]
    Window: 4
    [Calculated window size: 4]
    [Window size scaling factor: 1]
    Checksum: 0xe915 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    [Timestamps]
    [SEQ/ACK analysis]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
No.     Time           Source                Destination           Protocol Length Info
     15 0.000365700    127.0.0.1             127.0.0.1             TCP      44     55454 → 5001 [FIN, ACK] Seq=13 Ack=6 Win=3 Len=0
Frame 15: Packet, 44 bytes on wire (352 bits), 44 bytes captured (352 bits) on interface \Device\NPF_Loopback, id 0
Null/Loopback
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 55454, Dst Port: 5001, Seq: 13, Ack: 6, Len: 0
    Source Port: 55454
    Destination Port: 5001
    [Stream index: 0]
    [Stream Packet Number: 15]
    [Conversation completeness: Complete, WITH_DATA (31)]
```

```
    [TCP Segment Len: 0]
    Sequence Number: 13    (relative sequence number)
    Sequence Number (raw): 992947622
    [Next Sequence Number: 14    (relative sequence number)]
    Acknowledgment Number: 6    (relative ack number)
    Acknowledgment number (raw): 2008480772
    0101 .... = Header Length: 20 bytes (5)
    Flags: 0x011 (FIN, ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Accurate ECN: Not set
        .... 0... .... = Congestion Window Reduced: Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...1 .... = Acknowledgment: Set
        .... .... 0... = Push: Not set
        .... .... .0.. = Reset: Not set
        .... .... ..0. = Syn: Not set
        .... .... ...1 = Fin: Set
            [Expert Info (Chat/Sequence): Connection finish (FIN)]
                [Connection finish (FIN)]
                [Severity level: Chat]
                [Group: Sequence]
        [TCP Flags: ·······A···F]
            [Expert Info (Note/Sequence): This frame initiates the connection closing]
                [This frame initiates the connection closing]
                [Severity level: Note]
                [Group: Sequence]
    Window: 3
    [Calculated window size: 3]
    [Window size scaling factor: 1]
    Checksum: 0xe915 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    [Timestamps]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
No.     Time           Source                Destination           Protocol Length Info
     16 0.000374000    127.0.0.1             127.0.0.1             TCP      44     5001 → 55454 [ACK] Seq=6 Ack=14 Win=4 Len=0
Frame 16: Packet, 44 bytes on wire (352 bits), 44 bytes captured (352 bits) on interface \Device\NPF_Loopback, id 0
Null/Loopback
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 5001, Dst Port: 55454, Seq: 6, Ack: 14, Len: 0
    Source Port: 5001
    Destination Port: 55454
    [Stream index: 0]
    [Stream Packet Number: 16]
    [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 0]
    Sequence Number: 6    (relative sequence number)
    Sequence Number (raw): 2008480772
    [Next Sequence Number: 6    (relative sequence number)]
    Acknowledgment Number: 14    (relative ack number)
    Acknowledgment number (raw): 992947623
    0101 .... = Header Length: 20 bytes (5)
    Flags: 0x010 (ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Accurate ECN: Not set
        .... 0... .... = Congestion Window Reduced: Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...1 .... = Acknowledgment: Set
        .... .... 0... = Push: Not set
        .... .... .0.. = Reset: Not set
        .... .... ..0. = Syn: Not set
        .... .... ...0 = Fin: Not set
        [TCP Flags: ·······A····]
    Window: 4
    [Calculated window size: 4]
    [Window size scaling factor: 1]
    Checksum: 0xe914 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    [Timestamps]
    [SEQ/ACK analysis]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
No.     Time           Source                Destination           Protocol Length Info
```

```
     17 0.000411500    127.0.0.1             127.0.0.1             TCP      44     5001 → 55454 [FIN, ACK] Seq=6 Ack=14 Win=4 Len=0
Frame 17: Packet, 44 bytes on wire (352 bits), 44 bytes captured (352 bits) on interface \Device\NPF_Loopback, id 0
Null/Loopback
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 5001, Dst Port: 55454, Seq: 6, Ack: 14, Len: 0
    Source Port: 5001
    Destination Port: 55454
    [Stream index: 0]
    [Stream Packet Number: 17]
    [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 0]
    Sequence Number: 6    (relative sequence number)
    Sequence Number (raw): 2008480772
    [Next Sequence Number: 7    (relative sequence number)]
    Acknowledgment Number: 14    (relative ack number)
    Acknowledgment number (raw): 992947623
    0101 .... = Header Length: 20 bytes (5)
    Flags: 0x011 (FIN, ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Accurate ECN: Not set
        .... 0... .... = Congestion Window Reduced: Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...1 .... = Acknowledgment: Set
        .... .... 0... = Push: Not set
        .... .... .0.. = Reset: Not set
        .... .... ..0. = Syn: Not set
        .... .... ...1 = Fin: Set
            [Expert Info (Chat/Sequence): Connection finish (FIN)]
                [Connection finish (FIN)]
                [Severity level: Chat]
                [Group: Sequence]
        [TCP Flags: ·····A···F]
            [Expert Info (Note/Sequence): This frame undergoes the connection closing]
                [This frame undergoes the connection closing]
                [Severity level: Note]
                [Group: Sequence]
    Window: 4
    [Calculated window size: 4]
    [Window size scaling factor: 1]
    Checksum: 0xe913 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    [Timestamps]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
No.     Time           Source                Destination           Protocol Length Info
     18 0.000426000    127.0.0.1             127.0.0.1             TCP      44     55454 → 5001 [ACK] Seq=14 Ack=7 Win=3 Len=0
Frame 18: Packet, 44 bytes on wire (352 bits), 44 bytes captured (352 bits) on interface \Device\NPF_Loopback, id 0
Null/Loopback
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 55454, Dst Port: 5001, Seq: 14, Ack: 7, Len: 0
    Source Port: 55454
    Destination Port: 5001
    [Stream index: 0]
    [Stream Packet Number: 18]
    [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 0]
    Sequence Number: 14    (relative sequence number)
    Sequence Number (raw): 992947623
    [Next Sequence Number: 14    (relative sequence number)]
    Acknowledgment Number: 7    (relative ack number)
    Acknowledgment number (raw): 2008480773
    0101 .... = Header Length: 20 bytes (5)
    Flags: 0x010 (ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Accurate ECN: Not set
        .... 0... .... = Congestion Window Reduced: Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...1 .... = Acknowledgment: Set
        .... .... 0... = Push: Not set
        .... .... .0.. = Reset: Not set
        .... .... ..0. = Syn: Not set
        .... .... ...0 = Fin: Not set
        [TCP Flags: ·····A····]
    Window: 3
```

```
    [Calculated window size: 3]
    [Window size scaling factor: 1]
    Checksum: 0xe914 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    [Timestamps]
    [SEQ/ACK analysis]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
```